

Identification and Prioritization of Decision Criteria for Visual Style Selection in Contemporary Graphic Design Using the DEMATEL Method

Abstract

Purpose: Selecting an appropriate visual style is a fundamental decision in contemporary graphic design, as it shapes how visual messages are presented and perceived by audiences. Although this decision involves multiple decision criteria, their structural relationships have received limited systematic attention. Therefore, this study aims to identify and prioritize the key decision criteria for visual style selection in contemporary graphic design and to examine the structural relationships among these criteria using the DEMATEL method.

Methodology: This applied study adopted a descriptive–analytical approach. Eight decision criteria were identified through a review of the relevant literature and theoretical foundations. A DEMATEL-based pairwise comparison questionnaire was completed by 15 experts, including university professors, senior graphic designers, art directors, and experienced professionals in visual communication. The collected data were analyzed using the standard DEMATEL procedure to determine the prominence of each criterion and to reveal the underlying cause-and-effect relationships.

Findings: The analysis showed that Design Objectives and Communication Message, Designer Creativity and Experience, and Design Technology and Tools represent the primary causal criteria within the decision-making structure. In contrast, Target Audience, Cultural Identity, and Brand Identity and Client Requirements were identified as criteria mainly influenced by other decision criteria. Overall, the findings indicate that visual style selection is a multi-criteria decision-making process shaped by the interaction of interconnected decision criteria rather than isolated design considerations.

Originality/Value: This study applies the DEMATEL method to examine the structural relationships among decision criteria involved in visual style selection in contemporary graphic design. By providing a structured analysis of these relationships, the study contributes to a more systematic understanding of strategic decision-making in visual style selection.

Keywords: Graphic Design; Visual Style Selection; Strategic Decision-Making; DEMATEL; Multi-Criteria Decision Making.

Introduction

Contemporary graphic design has become an essential component of visual communication, playing a significant role in branding, marketing, information dissemination, and the communication of cultural and social values. Beyond aesthetic expression, graphic design serves as a strategic means of conveying messages and influencing audience perception. Consequently, the effectiveness of a design project depends not only on visual quality and technical execution but also on the extent to which design decisions correspond to communication objectives, audience expectations, and contextual requirements.

Among the various decisions made during the design process, selecting an appropriate visual style is one of the most important. Visual style shapes the overall identity of a design and determines how information is presented, interpreted, and remembered by audiences. Whether a project adopts a minimalist, modern, illustrative, typographic, or hybrid style, the final choice is rarely based on artistic preference alone. Instead, it reflects a deliberate decision-making

process in which multiple criteria must be evaluated simultaneously to achieve effective visual communication.

The selection of a visual style requires designers to balance a wide range of considerations, including communication objectives, audience characteristics, client requirements, cultural identity, available technologies, time and budget constraints, and their own professional knowledge and creative experience. These considerations do not operate independently; rather, they interact with one another throughout the design process. As a result, visual style selection can be regarded as a multi-criteria decision-making problem in which the relationships among decision criteria are as important as the criteria themselves.

Although visual style has attracted considerable attention in graphic design research, previous studies have primarily examined its aesthetic characteristics, its influence on audience perception, or the evaluation of completed design outcomes. Comparatively less attention has been devoted to understanding how designers make strategic decisions when selecting a visual style and how different decision criteria interact throughout this process. Most existing studies have focused on individual criteria rather than examining the structural relationships among them. Consequently, a systematic understanding of the interdependencies and causal relationships among decision criteria in visual style selection remains limited.

This issue has become increasingly important as organizations, brands, and creative industries operate in highly competitive and rapidly evolving communication environments. The expansion of digital media, diversification of audience expectations, and growing demand for consistent brand communication have increased the strategic importance of selecting an appropriate visual style. Decisions made at this stage may influence the clarity of communication, audience engagement, brand consistency, and the overall effectiveness of design projects. Therefore, identifying the key decision criteria and understanding their relationships can provide valuable guidance for designers, art directors, and creative managers.

Considering the interconnected nature of these decision criteria, conventional analytical approaches may not adequately explain the structure of the decision-making process. The Decision-Making Trial and Evaluation Laboratory (DEMATEL) method provides an appropriate analytical approach for examining complex relationships among decision criteria by distinguishing causal relationships from influenced relationships and revealing the overall structure of the decision-making system. This capability makes DEMATEL particularly suitable for investigating strategic decision-making in visual style selection.

Accordingly, this study aims to identify and prioritize the decision criteria involved in visual style selection in contemporary graphic design and to analyze the structural relationships among these criteria using the DEMATEL method. By providing a systematic analysis of these relationships, the study seeks to contribute to a better understanding of strategic decision-making in graphic design and to provide a structured analytical perspective for future research and professional practice.

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1.1|Research Question

How are the key decision criteria for visual style selection structured, and what causal relationships exist among these criteria in contemporary graphic design?

2|Theoretical Background

Contemporary graphic design has become an integral component of visual communication, supporting branding, marketing, information dissemination, education, and the communication of cultural and social values. With the rapid expansion of digital media and communication technologies, graphic design has evolved beyond a purely aesthetic activity into a strategic means of delivering messages and creating meaningful interactions between organizations and their audiences. Consequently, the effectiveness of a graphic design project depends not only on its visual quality but also on its ability to communicate intended messages and achieve specific communication objectives.

Visual style represents one of the most fundamental components of graphic design. It refers to the combination of formal, aesthetic, and expressive characteristics that define the visual identity of a design and shape how information is presented and perceived [1]. Elements such as color, typography, composition, illustration, visual hierarchy, and information organization collectively contribute to the formation of a visual style. Previous studies have shown that these visual characteristics can significantly influence audience attention, interpretation, and message comprehension. Therefore, selecting an appropriate visual style should be considered a strategic design decision rather than merely an aesthetic preference.

From a design management perspective, visual style selection represents a multi-criteria decision-making process. Designers are required to evaluate multiple decision criteria simultaneously, including communication objectives, audience characteristics, client requirements, cultural identity, technological capabilities, operational constraints, and their own professional knowledge and creative experience. Since these criteria interact throughout the design process, visual style selection cannot be adequately explained by considering each criterion independently. Instead, it should be viewed as a decision-making process in which multiple interconnected criteria collectively shape the final design outcome.

Previous studies have identified several important decision criteria associated with visual style selection. Design objectives and communication message provide the strategic direction of a project and guide subsequent design decisions. Target audience characteristics influence how visual information is interpreted, while cultural identity contributes to the appropriateness and contextual relevance of visual communication. Similarly, brand identity and client requirements establish practical and strategic boundaries that designers must consider throughout the design process.

Operational considerations also play an important role in visual style selection. Designer creativity and professional experience affect the evaluation of alternative design solutions, whereas time and budget constraints may restrict available options during project implementation. In addition, the presentation platform influences the suitability of different visual styles, as printed media, websites, mobile applications, and social media platforms require different design approaches. Technological advances and digital design tools have further expanded the range of available visual solutions, making technology an increasingly important consideration in contemporary graphic design practice.

Based on these theoretical foundations, visual style selection can be conceptualized as a multi-criteria decision-making process involving a network of interconnected decision criteria. This conceptual perspective provides the theoretical basis for applying the DEMATEL method to analyze the structural relationships among these criteria and to better understand the decision-making process underlying visual style selection in contemporary graphic design.

Bertão et al. (2023) investigated the capability of AI-powered logo makers to generate visually effective logo designs based on established visual design principles. Using a mixed-method approach, they examined whether AI-generated logos satisfied fundamental principles such as proportion, balance, and unity through expert evaluations. The findings indicated that although AI tools facilitate the logo design process, they remain limited in incorporating essential visual design principles and creative judgment. However, the study focused on evaluating AI-generated logo designs rather than examining the decision criteria involved in visual style selection.[2]

Ginosar and Zoran (2022) investigated the role of visual selection in the design process and its relationship with the formation of designers' personal styles. They developed a visual selection catalog within a parametric typography environment and conducted a user study involving graphic designers to analyze their visual selection patterns. The findings revealed that designers exhibit consistent and recurring selection behaviors, suggesting that visual selection reflects individual design styles. The study also demonstrated that style selection can be interpreted as a sequence of interconnected decision processes. However, the research focused primarily on visual selection in parametric typography and did not investigate the structural relationships among the decision criteria involved in visual style selection in contemporary graphic design [3].

Nair (2024) examined the impact of AI-generated imagery on the creative performance of design professionals and amateurs. Using surveys, practical design tasks, and interviews with participants from design and non-design backgrounds, the study investigated whether generative AI tools could substitute for human designers and how design education influences creative outcomes. The findings indicated that participants with formal design education produced more coherent, creative, and visually effective outputs than non-designers, highlighting the continuing importance of design knowledge and creative expertise despite the increasing adoption of AI tools. However, the study focused on the influence of AI on creative performance rather than on the decision criteria and structural relationships involved in visual style selection in contemporary graphic design[4]

Yan et al. (2024) investigated the application of artificial intelligence technology in visual communication design to improve design efficiency and foster innovation in visual style development. The study proposed an AI-based framework for the automatic arrangement of graphic elements and demonstrated that intelligent technologies can enhance visual communication performance, increase user satisfaction, and support the development of innovative visual design styles. However, the research primarily focused on algorithmic optimization and AI-assisted design generation, without examining the strategic decision criteria or the structural relationships underlying visual style selection in contemporary graphic design.[5]

Huang et al. (2023) presented a comprehensive survey of graphic design intelligence by systematically reviewing research on design representation, design understanding, and design generation. The authors emphasized that graphic design is fundamentally a visual communication process that requires the careful selection and integration of visual elements, such as color, typography, shape, and layout, in accordance with design principles and communication objectives. They also highlighted the growing role of artificial intelligence in supporting designers by automating repetitive tasks while preserving the importance of human creativity and professional judgment. However, the study provided a conceptual overview of graphic design intelligence and did not investigate the structural relationships among the decision criteria involved in visual style selection.[6]

3|Conceptual Model of the Research

Based on the theoretical foundations and literature review, visual style selection in contemporary graphic design is conceptualized as a multi-criteria decision-making process influenced by a set of communication, contextual, managerial, operational, and technological decision criteria. Accordingly, eight key decision criteria were identified: Design Objectives and Communication Message (C1), Target Audience (C2), Cultural Identity (C3), Brand Identity and Client Requirements (C4), Designer Creativity and Experience (C5), Time and Budget Constraints (C6), Presentation Platform (C7), and Design Technology and Tools (C8).

These decision criteria do not operate independently. Instead, they form an interconnected network in which each criterion may directly or indirectly influence other criteria within the decision-making process. For instance, brand identity and client requirements may shape communication objectives, while technological capabilities and presentation platforms may influence the designer's creative opportunities and the implementation of specific visual styles. Likewise, audience characteristics and cultural identity contribute to determining the appropriateness of different visual style alternatives.

Accordingly, the conceptual model of this study was developed as a network-based structure in which visual style selection occupies the central position and is influenced by the interactions among the identified decision criteria. The model assumes that strategic decision-making is determined not only by the relative importance of individual criteria but also by the causal relationships among them. To examine these structural relationships, the Decision-Making Trial and Evaluation Laboratory (DEMATEL) method was employed. This method enables the identification of cause and effect criteria and provides a systematic understanding of the structural relationships governing visual style selection in contemporary graphic design.

Table 1. Decision Criteria for Visual Style Selection in Contemporary Graphic Design

Code	Criterion
C1	Design Objective and Message
C2	Target Audience
C3	Cultural Identity
C4	Brand Identity and Client Requirements
C5	Designer Creativity and Experience
C6	Time and Budget Constraints
C7	Presentation Context (Media Platform)
C8	Design Technology and Tools

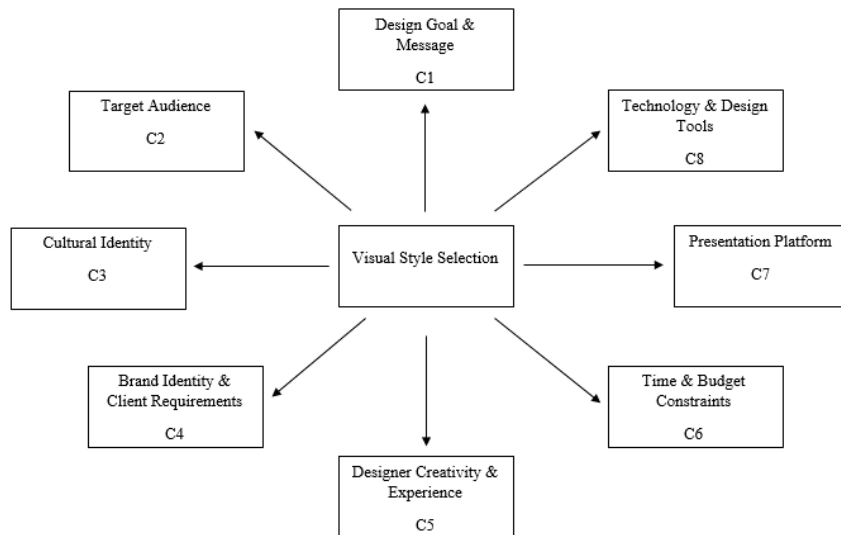


Figure 1. Conceptual Model of Strategic Decision-Making for Visual Style Selection in Contemporary Graphic Design

4|Research Methodology

This study is applied in terms of its purpose and adopts a descriptive–analytical research design. Its primary objective is to identify and analyze the decision criteria involved in strategic visual style selection in contemporary graphic design and to examine the structural relationships among these criteria using the Decision-Making Trial and Evaluation Laboratory (DEMATEL) method.

The research process was conducted in three stages. In the first stage, the decision criteria involved in visual style selection were identified through a comprehensive review of the relevant literature, previous studies, and theoretical foundations related to graphic design, visual communication, and decision-making. In the second stage, a DEMATEL-based pairwise comparison questionnaire was developed and distributed among experts in the field of graphic design. In the third stage, the collected data were aggregated and analyzed following the standard DEMATEL procedure, including the construction of the direct-relation matrix, matrix normalization, calculation of the total-relation matrix, and extraction of prominence and relation indices. The study population consisted of university professors, art directors, graphic designers, and professionals with practical experience in visual design. A purposive sampling approach was adopted, and fifteen experts with relevant academic and professional backgrounds in graphic design participated in the study. The experts were selected based on their professional experience, practical expertise, and familiarity with graphic design and visual decision-making processes.

4.1|Data Collection Instrument

Data were collected using a DEMATEL-based pairwise comparison questionnaire. In this questionnaire, the experts evaluated the degree of influence of each decision criterion on every other criterion using a five-point scale ranging from 0 to 4, where 0 represented No Influence, 1 indicated Low Influence, 2 represented Moderate Influence, 3 indicated High Influence, and 4 represented Very High Influence. The questionnaire was developed based on the eight decision criteria identified through the literature review: Design Objectives and Communication Message (C1), Target Audience (C2), Cultural Identity (C3), Brand Identity and Client Requirements (C4), Designer Creativity and Experience (C5), Time and Budget Constraints (C6), Presentation Platform (C7), and Design Technology and Tools (C8). The experts evaluated the direct influence of each decision criterion on every other criterion through pairwise comparisons.

4.2 |DEMATEL Procedure

Data analysis was conducted according to the standard DEMATEL procedure. First, the average scores provided by the experts were calculated to construct the direct-relation matrix, representing the direct influence of each decision criterion on the others. Next, the matrix was normalized to ensure that all values fell within a standardized range. Based on the normalized matrix, the total-relation matrix was then calculated to represent both direct and indirect relationships among the decision criteria.

Subsequently, the row sums (D) and column sums (R) were calculated for each criterion. The index (D + R) represents the prominence of a criterion within the decision-making system, while (D – R) indicates its causal role. Criteria with positive (D – R) values were classified as cause criteria, whereas those with negative values were classified as effect criteria.

Finally, a cause–effect diagram was generated to visualize the structural relationships among the identified decision criteria and to facilitate the interpretation of the strategic decision-making structure underlying visual style selection in contemporary graphic design.

5|Research Findings

To analyze the structural relationships among the decision criteria involved in strategic visual style selection, the data collected from the expert questionnaires were analyzed using the DEMATEL method. The findings obtained from each stage of the analysis are presented below.

5.1|Construction of the Direct-Relation Matrix

Following the collection of expert evaluations, the average scores assigned to the relationships among the decision criteria were calculated, and the direct-relation matrix was constructed. As presented in **Table 2**, this matrix represents the degree of direct influence of each decision criterion on the others.

Table 2. Direct-Relation Matrix of the Decision Criteria

	C1	C2	C3	C4	C5	C6	C7	C8
C1	0	3	2	3	2	2	2	1
C2	2	0	2	2	1	1	3	1
C3	1	2	0	2	1	1	2	1
C4	3	2	2	0	2	2	2	1
C5	2	1	1	2	0	2	2	3
C6	1	1	1	2	1	0	2	2
C7	2	3	2	2	2	2	0	2
C8	2	1	1	2	3	2	2	0

5.2|Normalized Direct-Relation Matrix

Following the construction of the direct-relation matrix, the values were normalized to ensure comparability among the decision criteria and to satisfy the requirements of the DEMATEL methodology. The normalization process transformed the original values into a standardized scale while preserving the relative influence among the decision criteria. As presented in **Table 3**, the normalized direct-relation matrix serves as the basis for calculating the total-relation matrix and represents the standardized levels of direct influence among the identified decision criteria.

Table 3. Normalized Direct-Relation Matrix.

	C1	C2	C3	C4	C5	C6	C7	C8
C1	0.00	0.19	0.13	0.19	0.13	0.13	0.13	0.06
C2	0.13	0.00	0.13	0.13	0.06	0.06	0.19	0.06
C3	0.06	0.13	0.00	0.13	0.06	0.06	0.13	0.06
C4	0.19	0.13	0.13	0.00	0.13	0.13	0.13	0.06
C5	0.13	0.06	0.06	0.13	0.00	0.13	0.13	0.19
C6	0.06	0.06	0.06	0.13	0.06	0.00	0.13	0.13
C7	0.13	0.19	0.13	0.13	0.13	0.13	0.00	0.13
C8	0.13	0.06	0.06	0.13	0.19	0.13	0.13	0.00

5.3|Total-Relation Matrix

Based on the normalized direct-relation matrix, the total-relation matrix was calculated to determine both the direct and indirect relationships among the decision criteria. Unlike the direct-relation matrix, the total-relation matrix captures not only the immediate influence of one criterion on another but also the indirect influences transmitted through other interconnected criteria within the decision-making system.

As presented in **Table 4**, the total-relation matrix provides a comprehensive representation of the interaction structure among the decision criteria involved in visual style selection. The values reported in this matrix form the basis for calculating the prominence and relation indices used in the subsequent stage of the DEMATEL analysis.

Table 4. Total-Relation Matrix.

	C1	C2	C3	C4	C5	C6	C7	C8
C1	0.21	0.34	0.25	0.36	0.28	0.26	0.32	0.22
C2	0.24	0.18	0.23	0.25	0.19	0.18	0.30	0.20
C3	0.18	0.22	0.15	0.24	0.17	0.16	0.23	0.17
C4	0.29	0.30	0.27	0.21	0.26	0.24	0.28	0.21
C5	0.27	0.20	0.19	0.27	0.20	0.25	0.29	0.33
C6	0.19	0.17	0.16	0.24	0.18	0.15	0.23	0.22
C7	0.28	0.31	0.26	0.30	0.27	0.25	0.19	0.27
C8	0.26	0.19	0.18	0.25	0.31	0.23	0.26	0.18

5.4|Analysis of Prominence and Causal Relationships

To determine the structural role of each decision criterion within the decision-making system, the values of D (sum of rows), R (sum of columns), D + R (prominence), and D – R (relation) were calculated. As shown in **Table 5**, the (D + R) index indicates the overall prominence and level of interaction of each decision criterion within the system, whereas the (D – R) index determines whether a criterion belongs to the cause or effect group.

Decision criteria with positive (D – R) values were classified as cause criteria, indicating that they exert greater influence on other criteria than they receive. Conversely, criteria with negative (D – R) values were classified as effect criteria, indicating that they are primarily influenced by other criteria within the system.

Table 5. Values of D, R, D+R, and D–R.

Criteria	D	R	D+R	D–R	Role
C1: Design Objectives and Communication Message	2.17	1.92	4.09	0.25	Cause
C2: Target Audience	1.77	1.91	3.68	-0.14	Effect
C3: Cultural Identity	1.58	1.69	3.27	-0.11	Effect
C4: Brand Identity and Client Requirements	2.05	2.22	4.27	-0.17	Effect
C5: Designer Creativity and Experience	2.23	2.01	4.24	0.22	Cause
C6: Time and Budget Constraints	1.52	1.67	3.19	-0.15	Effect
C7: Presentation Platform	2.19	2.18	4.37	0.01	Cause
C8: Design Technology and Tools	2.21	2.13	4.34	0.08	Cause

According to **Table 5**, Design Objectives and Communication Message (C1) emerged as the most influential cause criterion, followed by Designer Creativity and Experience (C5) and Design Technology and Tools (C8). These criteria play a central role in shaping strategic visual style selection by exerting substantial influence on the remaining decision criteria.

In contrast, Target Audience (C2), Cultural Identity (C3), Brand Identity and Client Requirements (C4), and Time and Budget Constraints (C6) were identified as effect criteria, indicating that they are primarily influenced by other decision criteria within the system. Furthermore, the prominence values (D + R) revealed that Presentation Platform (C7), Design Technology and Tools (C8), and Brand Identity and Client Requirements (C4) exhibit the highest levels of interaction within the decision-making network, highlighting their strategic importance in visual style selection.

6|Discussion

The findings of this study demonstrate that visual style selection in contemporary graphic design should be understood as a strategic multi-criteria decision-making process rather than as a purely aesthetic activity. By applying the DEMATEL method, this study revealed not only the relative importance of the identified decision criteria but also the structural relationships through which these criteria influence one another. This perspective extends previous research by emphasizing that effective visual style selection depends on the interaction of multiple interconnected criteria rather than on isolated design considerations.

Among the identified decision criteria, Design Objectives and Communication Message (C1) emerged as the most influential cause criterion. This finding suggests that communication objectives constitute the primary foundation for visual style selection. Rather than selecting a visual style based solely on aesthetic preferences, designers first establish the intended purpose and communication goals of a project, which subsequently shape other design decisions. This result is consistent with design management literature, which considers communication effectiveness as the central objective of professional graphic design practice.

The findings also highlight the significant role of Designer Creativity and Experience (C5). Although digital technologies have transformed contemporary design practice, the results indicate that strategic design decisions continue to rely heavily on designers' professional expertise, creative thinking, and

practical experience. This suggests that technological development complements rather than replaces human creativity in the decision-making process.

Another important finding concerns the role of Design Technology and Tools (C8) as a cause criterion. The continuous evolution of digital design software, artificial intelligence-supported design applications, and interactive media has considerably expanded designers' creative possibilities. Consequently, technology has become an important enabling element that influences both the feasibility of design alternatives and the implementation of visual styles within contemporary graphic design projects.

In contrast, Target Audience (C2), Cultural Identity (C3), Brand Identity and Client Requirements (C4), and Time and Budget Constraints (C6) were identified as effect criteria. This does not imply that these criteria are less important. Instead, the findings indicate that their roles are shaped by the broader strategic decisions established during earlier stages of the design process. In other words, contextual considerations become meaningful after communication objectives, creative direction, and technological capabilities have defined the overall strategic framework.

Overall, the results suggest that strategic decision-making in visual style selection is governed by an interconnected system of decision criteria in which causal relationships play a more important role than isolated individual criteria. Understanding these structural relationships can assist designers and creative managers in developing more systematic and informed approaches to visual style selection while providing a useful analytical perspective for future research in graphic design and design management.

7. Conclusion

The primary objective of this study was to identify and prioritize the decision criteria involved in strategic visual style selection in contemporary graphic design and to examine the structural relationships among these criteria using the DEMATEL method. The findings demonstrate that visual style selection is a complex multi-criteria decision-making process in which interconnected decision criteria collectively shape strategic design decisions rather than operating independently.

Among the identified decision criteria, Design Objectives and Communication Message (C1), Designer Creativity and Experience (C5), and Design Technology and Tools (C8) were identified as the principal cause criteria, indicating that they play leading roles in influencing other criteria within the decision-making system. In contrast, Target Audience (C2), Cultural Identity (C3), Brand Identity and Client Requirements (C4), and Time and Budget Constraints (C6) functioned primarily as effect criteria, suggesting that their roles are influenced by broader strategic decisions established during the design process.

From a theoretical perspective, this study contributes to the growing body of knowledge on graphic design and design management by conceptualizing visual style selection as a structured multi-criteria decision-making process and by revealing the causal relationships among the identified decision criteria. From a practical perspective, the findings provide designers, art directors, and creative managers with a systematic framework for understanding how different decision criteria interact during visual style selection, thereby supporting more informed and strategically aligned design decisions.

Overall, this study demonstrates that effective visual style selection extends beyond aesthetic judgment and requires a structured evaluation of multiple interconnected decision criteria. The proposed framework offers a foundation for future studies seeking to investigate strategic decision-making in graphic design and related creative disciplines.

7.1 Limitations and Future Research

This study has several limitations that should be considered when interpreting its findings. First, the analysis was based on the judgments of a relatively small group of experts with professional experience in graphic design, which may limit the generalizability of the results to other design contexts. Second, the study employed the DEMATEL method as the sole analytical technique; therefore, the identified relationships reflect the assumptions and characteristics of this method.

Future research may expand the proposed framework by including additional decision criteria or by examining visual style selection in different areas of design, such as digital media, user experience (UX) design, interactive design, and motion graphics. Researchers may also employ hybrid multi-criteria decision-making approaches, including DEMATEL–ANP, Fuzzy DEMATEL, or other advanced methods, to validate and extend the structural relationships identified in this study. In addition, comparative studies involving different industries, cultural contexts, or professional groups could provide a broader understanding of strategic decision-making in visual style selection.